MD Saib Hossain

AI Researcher | Machine Learning Engineer Dhaka, Bangladesh |
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Profile

AI Engineer and Machine Learning developer with hands-on experience in AI automation workflows, LLM-based systems, RAG pipelines, and computer vision models. Proven record of building production-style AI microservices using Python and FastAPI, integrating ML models with APIs and databases, and developing multi-agent AI systems. Research-active with published work in computer vision and medical imaging. Strong focus on scalable AI solutions, automation pipelines, and real-world deployment.

Experience

1. Machine Learning Engineer Intern | Fluvo Soft (Remote) | *Nov 2025 - Mar 2026*

a. Designed a synthetic face generation pipeline with StyleGAN2 and Stable Diffusion XL (SDXL) to address data scarcity. Fine-tuned diffusion models for identity preservation and controlled diversity.
b. Applied Adaptive Discriminator Augmentation (ADA) to mitigate overfitting with small datasets.
c. Conducted error analysis and quality evaluation to improve generative consistency and realism.
d. Developed an agentic RAG pipeline using LangChain/LangGraph, enabling automated workflow orchestration and intelligent decision support.

2. Research Assistant | *Institute of Advanced Research (IAR), United International University (UIU)* | *Nov 2025 - Present*

a. Developed deep learning pipelines for medical image analysis.
b. Extracted and analyzed radiomic features; supported model evaluation and statistical correlation studies.
c. Documented experiments and findings of model reliability, data bias, and clinical applicability, relevant to healthcare AI governance.

3. Volunteer Research Assistant | Ruach Tech (London, UK- Remote) *Jan - Mar 2026*

a. Contributing to the Engineering & Research Division, focusing on the theoretical underpinnings of safe Artificial Intelligence systems.
b. Conducting critical literature reviews and drafting research briefs to support exploratory research into advanced model capabilities and alignment.
c. Synthesizing complex technical papers into actionable insights for ongoing R&D projects.

Leadership & social work

General Secretary | National Children's Task Force (NCTF) Satkhira *2014 – 2017* Link

a. Led district-level operations for a nationwide organization supported by Save the Children and Plan International, focused on monitoring child rights implementation.
b. Conducted grassroots monitoring of the National Plan of Action against Sexual Abuse and Exploitation, contributing evidence to national stakeholders.
c. Worked at the intersection of policy frameworks, local governance, and real-world outcomes, translating field-level data into policy-relevant insights.
d. Facilitated coordination between child-led committees and local government bodies to advocate for stronger child protection policies.

Volunteer / Social Worker | Alo Bhubon Trust *(2025 – Present)* *Link*

a) Community Development & Climate Action: Facilitated community-driven initiatives focusing on rural health, education, and climate resilience. Mobilized resources and advocacy to deliver essential support to vulnerable populations in Bangladesh.
b) Mobilized resources and advocacy to assist vulnerable populations, reinforcing a commitment to socially responsible technology and policy engagement.

Projects

Agentic AI Technical Interviewer
Engineered a non-linear, autonomous AI interviewing agent that dynamically parses PDF resumes to generate tailored technical assessments. Orchestrated complex, stateful conversation flows including real-time answer evaluation, clarifying hints, and adaptive follow-ups using a state machine architecture, with seamless toggling between cloud (Google Gemini) and local (Ollama) LLM inference. **Tech use: LangGraph, LangChain, Gemini API, Ollama, Pydantic, Streamlit.** ***Link***

MedAssist-LLM-iPhone
Fine-tuned a lightweight ***Qwen2.5-0.5B-Instruct LLM*** on a medical Question-Answering dataset and optimized it for fully offline inference on an **iPhone device**. Implemented on-device deployment using quantization and mobile inference runtime to enable privacy-preserving medical QA without cloud dependency. ***Tech use: Unsloth, Hugging Face Transformers, LoRA Fine-Tuning, GGUF & llama.cpp, pocketpal. Link***

Real-Time Vision + LLM Agent "LLM with Eyes."
Developed an autonomous vision agent using the YOLO11 and Gemini 2.5 Flash LLM. Pipeline segments objects from a live camera feed in real-time and routes visual data to an LLM for context-aware identification.
Tech use: **Python, YOLO11, Gemini API.** ***Link***

Synthetic Face Dataset Generation (GANs & Diffusion)
Built a hybrid generative pipeline using **StyleGAN2-ADA + SDXL** to find how the dataset affects the face generation model. Applied **LoRA fine-tuning** for identity control and dataset consistency. Focused on **vision dataset engineering** for small-data research scenarios. ***Tech use: Python, PyTorch, GAN, diffusion model, and finetune.*** **Link**

LangGraph MCP Agent with XAMPP Database Integration
Developed a modular LangGraph-powered MCP server integrated with XAMPP's MySQL database for intelligent file uploads, metadata storage, and AI-driven query workflows. Designed for scalability and extendable into multi-agent architectures.
Tech use: Python 3.12, LangGraph, FastAPI, MCP Server, MySQL (XAMPP), Uvicorn, MySQL Connector. Link

Bangla Sign Language Translator
Built a vision-based system using MediaPipe and PyTorch that detects hand signs to predict Bangla alphabets and words. ***Technologies:- PyTorch, MediaPipe, Streamlit, gTTS, ESP32, Raspberry Pi. Link***

Googler-Search-Agent
Built a local AI agent that autonomously searches the web, retrieves results, and generates concise summaries using LangGraph and Ollama. Designed for offline-friendly environments with transparent query-to-summary workflows. ***Technologies: Python3, LangGraph, Web Automation, DuckDuckGo Search.*** -***Link***

Prompt-Based Dynamic LLM Selector
Developed a Python tool that intelligently routes prompts to locally hosted language models of varying sizes via Ollama, optimizing performance and resource usage. Designed for flexibility in multi-model environments with dynamic selection logic. ***Technologies: Python, Ollama, gemma3n:4B, deepseek-r1:1.5b, gemma 3:1b.*** -**Link**

DeepLearning-with-PyTorch
Implemented CNNs and optimization techniques for image classification on MNIST and FashionMNIST.
Technologies: PyTorch, Google Colab, Python. Link
AI-Based Plagiarism Detection

Implemented a lightweight system to detect AI-generated text in documents using on-device LLMs.
Technologies: Python, On-device LLM, NLP, Text Classification. -Link

AI-Powered Study Assistant Chatbot (GPT-2, FastAPI, JS)
Built an English chatbot using GPT-2 with FastAPI backend and JavaScript frontend for educational help.
Technologies: GPT-2, FastAPI, Python, JavaScript, HTML/CSS. Link

Publications (First Author)

1. "Unifying Adult and Pediatric Domains: Multi-Dataset Deep Learning for Fracture Classification, Detection, and Segmentation." *IEEE*
28th International Conference on Computer and Information Technology (ICCIT), 2025
Developed a unified training pipeline harmonizing FracAtlas (adult) and GRAZPEDWRI-DX (pediatric) datasets.
Improved cross-domain generalization, achieving gains in Classification AUC (+3.8%), Detection mAP (+4.2%), and Segmentation Dice (+2.7%).
Models: ResNet-50, YOLOv8/v11, and DeepLabV3.

2. "BanglaSign-Tutor: An interactive Bangla Sign Language Learning System via Finger-Guided Feedback." *IEEE ICCIT 2025*
Combining Computer vision with the LLM (Gemma3) pipeline for hand-sign recognition with BDSL.
Achieved 98.2% accuracy for alphabets and 99.5% for numerals.
User studies showed a 56.43% reduction in learning latency via real-time haptic and visual feedback.

3. "Urban Sustainability in Bangladesh: Satellite-Based Analysis of Land Use and Air Pollution Trends."*IEEE ICCIT 2025*
Analyzed Sentinel-2 & Sentinel-5 imagery using Google Earth Engine for (LULC) and pollution from 2019–2025.
Compared six ML algorithms (Random Forest, XGBoost, etc.).

Achievements & Awards

1st Runner-up, *Deep Learning in Computational and Spatial Biology Kaggle Competition* — Recognized for developing a **multi-semantic graph learning framework** leveraging spatial, transcriptomic, and morphological features for unsupervised cell-type discovery in spatial transcriptomics data.

Best Abstract Award, same competition

Technical Skills

Programming & Frameworks: Python, FastAPI, PyTorch, TensorFlow, Scikit-learn, OpenCV
LLM & RAG Systems: LLM/SLM, Fine-tuning (LoRA/QLoRA), RAG, Agentic RAG, Graph RAG, Semantic Search, Prompt Engineering
Agentic AI Tech: LangChain, LangGraph, agent workflows, Multi-agent systems, API-based AI integration.
Computer Vision: Object detection (YOLO), segmentation, medical imaging, dataset curation
Model Deployment & MLOps: Fast APIs, HuggingFace, Langsmith, mlflow, Subabase, Vercel.
Databases: MySQL, SQL, vector-style retrieval workflows, embedding pipelines, Fassi
Tools: Git, Ollama, Groq API, Streamlit, Uvicorn

Education

United International University, DHAKA — *BSCSE*
February 2022 - Current (4th year)

Additional Strengths

- Strong scientific writing and research presentation skills
- Strategic planning and communication

Certifications

Python and SQL -Link
Pytorch - Link
IBM AI Engineering - Link